DeepMind

Do language models form grounded conceptual spaces?

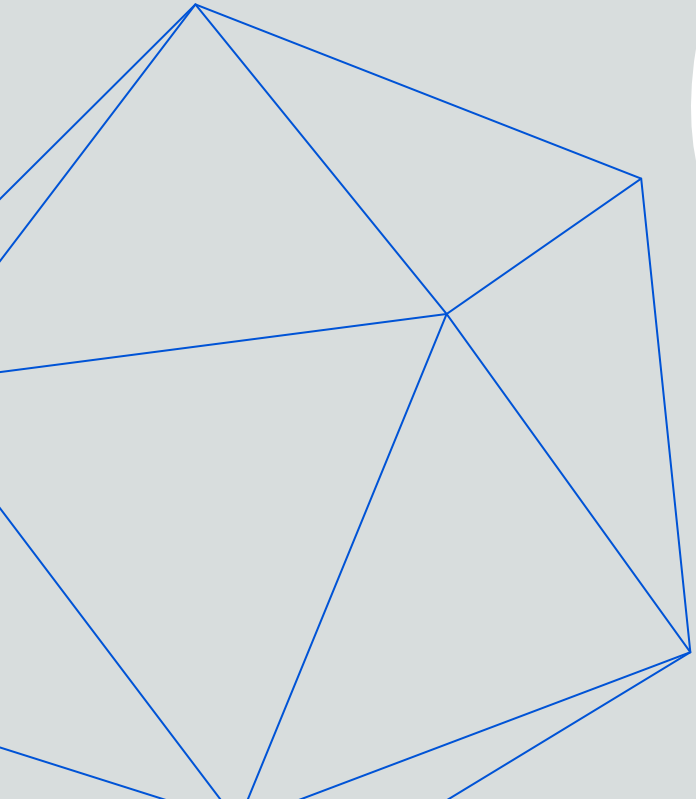
Roma Patel

October 2023
MILA Seminar on Multi-Agent Reinforcement Learning @ McGill University



DeepMind

The current state of large language models



The current state of language models

Private & Confidential







The current state of language models

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Can generate fluent, coherent and human-like text (QA, summarisation)

```
Title: United Methodists Agree to Historic Split Subtitle: Those who oppose gay marriage will form their own denomination Article: After two days of intense debate, the United Methodist Church has agreed to a historic split - one that is expected to end in the creation of a new denomination, one that will be "theologically and socially conservative," according to The Washington Post. The majority of delegates attending the church's annual General Conference in May voted to strengthen a ban on the ordination of LGBTQ clergy and to write new rules that will "discipline" clergy who officiate at same-sex weddings.
```

Language Models are Few-Shot Learners. Brown et. al., NeuRIPs 2020.

Can few-shot learn NLU tasks: entailment, sentiment, grammatical corrections

```
Poor English input:  The patient was died.
Good English output:  The patient died.

Poor English input:  We think that Leslie likes ourselves.
Good English output:  We think that Leslie likes us.

Poor English input:  Janet broke Bill on the finger.
Good English output:  Janet broke Bill's finger.

Poor English input:  Mary arranged for, in St.  Louis, John to rent a house cheap.
Good English output:  Mary arranged for John to rent a house in St.  Louis.
```

Context →	The bet, which won him dinner for four, was regarding the existence and mass of the top quark, an elementary particle discovered in 1995. question: The Top Quark is the last of six flavors of quarks predicted by the standard model theory of particle physics. True or False? answer:
Target Completion →	False

Figure G.31: Formatted dataset example for RTE

GLUE: A Multi-task Benchmark and Analysis Platform for Natural Language Understanding. Wang et. al., ICLR 2019.

The current state of language models

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Can reconstruct WordNet structure and learn all word relations!

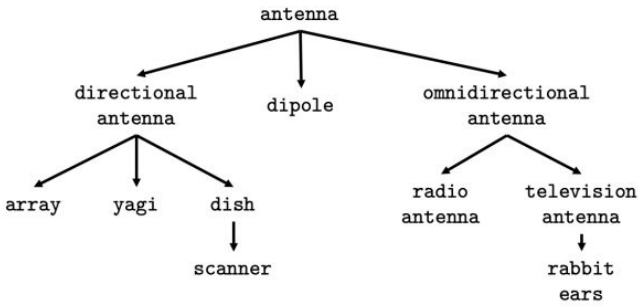
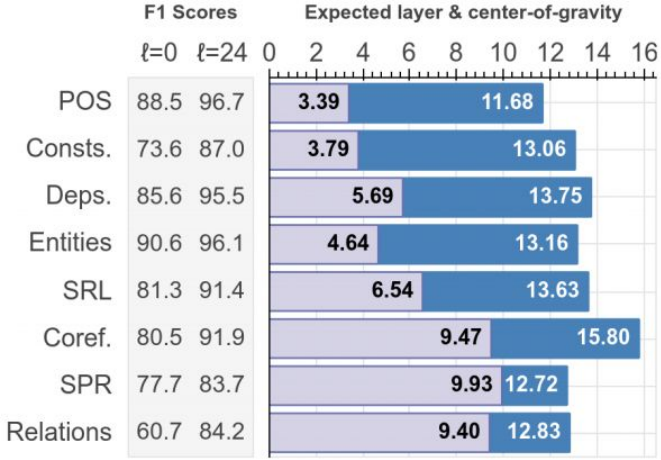


Figure 1: *An example subtree from the* WORDNET *hierarchy.*

Constructing Taxonomies from Pretrained Language Models. Chen, Lin and Klein. NAACL 2021.

Can solve most classical NLP tasks!



BERT Rediscovers the Classical NLP Pipeline. Tenney, Das and Pavlick. ACL 2019.

The current state of language models

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"We argue that the language modeling task, because it only uses form as training data, cannot in principle lead to learning of meaning".

[Bender and Koller 2020]



The current state of language models

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"We argue that the language modeling task, because it only uses form as training data, cannot in principle lead to learning of meaning".

[Bender and Koller 2020]



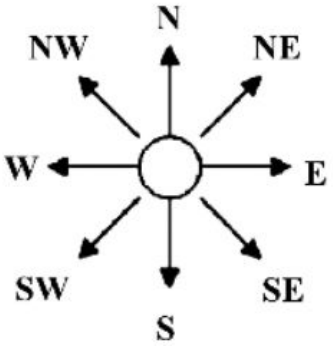
How can a model know what the words blue and green mean by only seeing the word forms?



blue



green



north

south

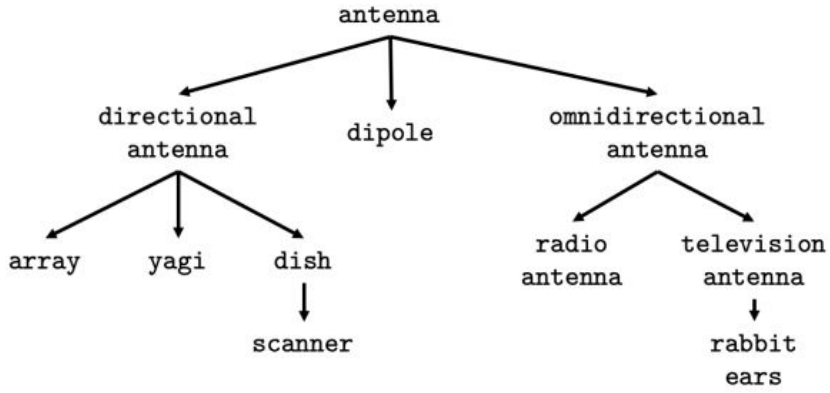
The current state of language models

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"We argue that the language modeling task, because it only uses form as training data, cannot in principle lead to learning of meaning".

[Bender and Koller 2020]

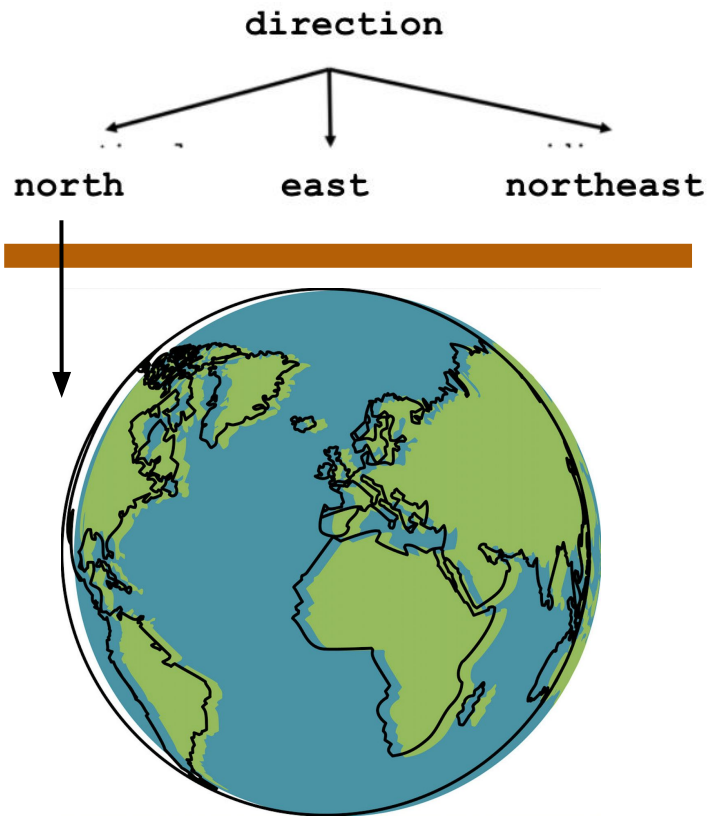


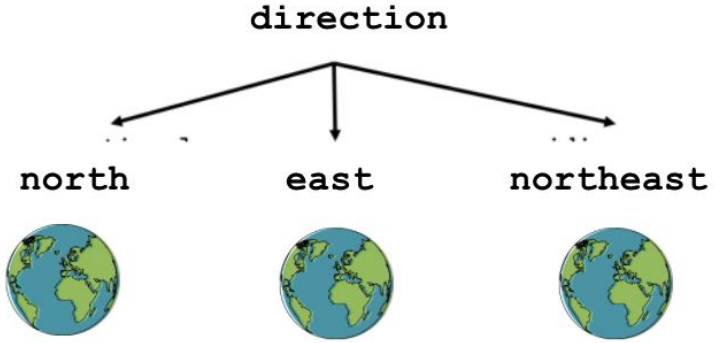


... but we do know that models at least learn relational structure between words

Mapping a language model's space to a grounded space

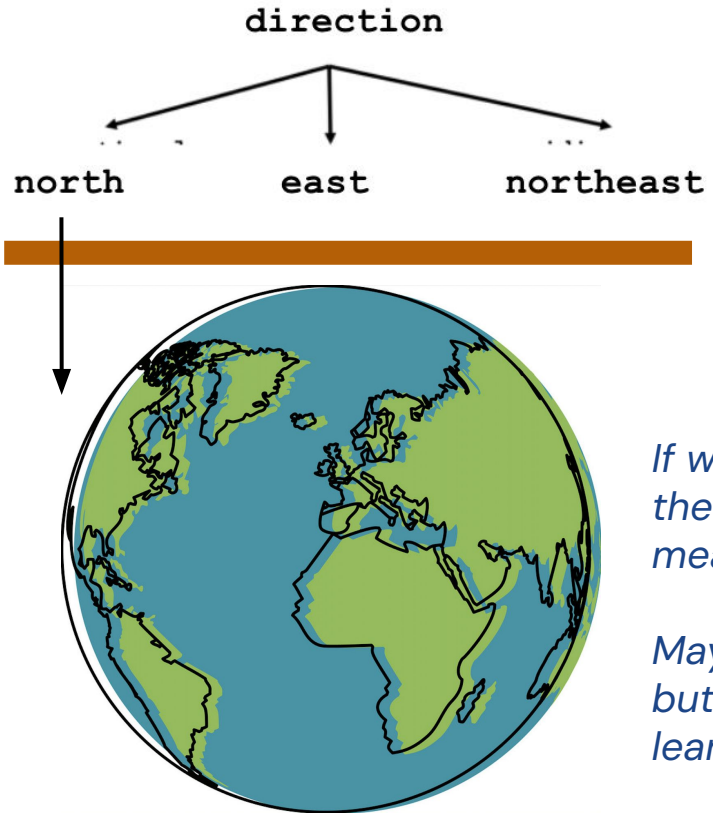
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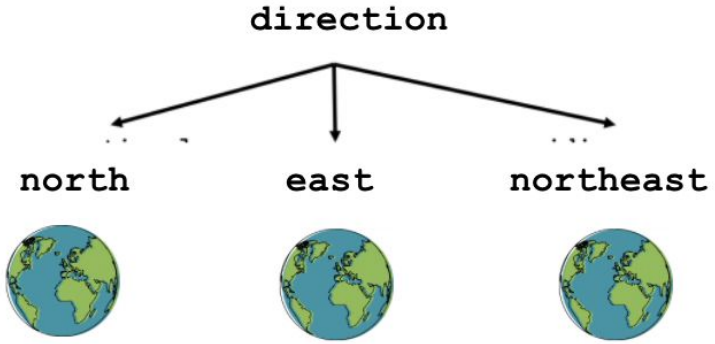




Mapping a language model's space to a grounded space

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*If we teach models how to ground **some** concepts, can they use the structure of language to extrapolate meaning to **all** related concepts?*

Maybe the text-only representation is not grounded but is it functionally equivalent to what would be learned if it was grounded?

Quinian bootstrapping & learning with placeholder symbols

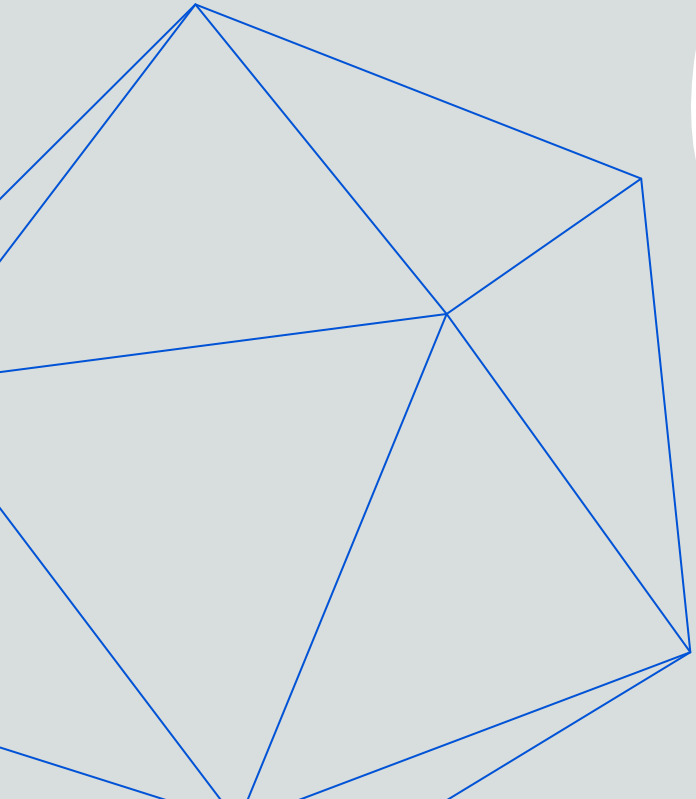
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(from Carey's "The Origin of Concepts")

1. **Relations among some symbols are learned directly, in terms of each other** ("proton", "mole")
2. Symbols are initially at most only partly interpreted in terms of antecedently available concepts
3. Symbols serve as **placeholders**
4. Modeling processes–**analogy**, inductive inference, thought experiment, limiting case analyses–are used to provide conceptual underpinnings for the placeholders
5. These modeling processes combine and integrate separate representations from distinct domain-specific conceptual systems
6. These processes **create explicit representations** of knowledge previously embodied in constraints on the computations



DeepMind



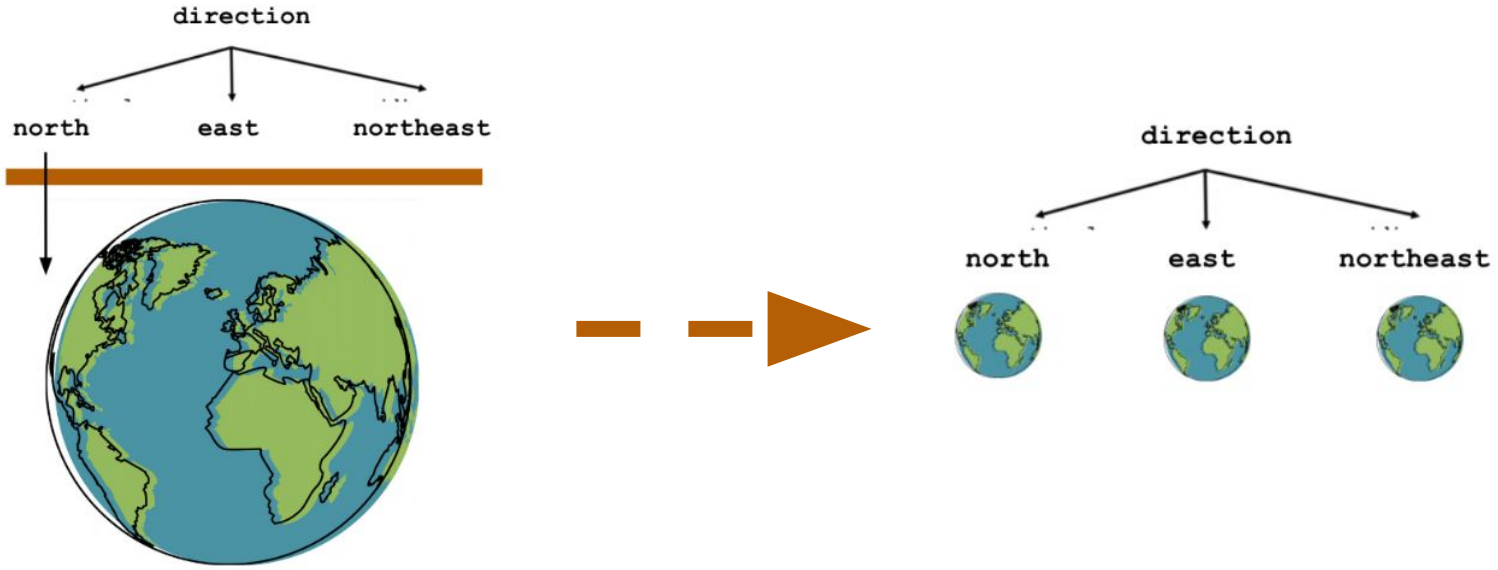
Part 1

Mapping Language Models to Grounded Conceptual Spaces

Motivation

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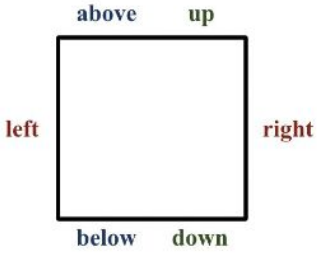
*Can we teach a text-only LLM to ground **some** concepts and test if it can extend this by analogy & relational structure to the remaining concepts (previous placeholder symbols)?*



Categories of concepts

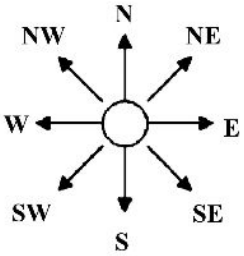
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Spatial Terms



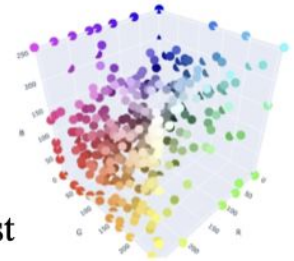
World:
[0. 0. 0. 0. 0.]
[0. 0. 0. **1**. 0.]
[0. 0. 0. 0. 0.]
Answer: right

Cardinal Directions



World:
[0. 0. **1**.]
[0. 0. 0.]
[0. 0. 0.]
Answer: northeast

RGB Colours



RGB: (255, 0 0)
Answer: red

RGB: (143, 0, 255)
Answer: violet

[Kim et. al., 2020, Monroe et. al., 2017]

Categories of concepts

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Spatial Terms

Example Input (20 in-context-learning examples followed by prompt)

```
World:
[0. 0. 0.]
[0. 0. 0.]
[0. 0. 1.]
Answer: right

World:
[1. 0. 0. 0.]
Answer: left

...13 more...

World:
[0. 0. 0.]
[0. 0. 0.]
[0. 0. 1.]
Answer: right

World:
[0. 0.]
[1. 0.]
[0. 0.]
Answer: left

World:
[0. 0. 0.]
[0. 0. 0.]
[0. 0. 1.]
[0. 0. 0.]
[0. 0. 0.]
Answer: right

World:
[0. 1. 0. 0.]
Answer: left

World:
[1. 0.]
[0. 0.]
[0. 0.]
[0. 0.]
[0. 0.]
Answer: left

World:
[1. 0. 0. 0.]
[0. 0. 0. 0.]
Answer:
```

Example Model Outputs

GPT-2 (124M)

world	P=0.09
0. 0.]]	P=0.08
[0 [0	P=0.01

GPT-3 (175B)

left	P=0.20
right	P=0.11
leftmost	P=0.01

Categories of concepts

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RGB Colours

Example Input (60 in-context-learning examples followed by prompt)

RGB: (48, 213, 200)
Answer: orange

RGB: (220, 20, 60)
Answer: crimson

RGB: (0, 0, 128)
Answer:

All 60 Training Examples



6 Primary and Secondary Colours

red, blue, yellow, green, orange, violet

57 Colours Within a Sub-space

dark red, maroon, crimson, fuchsia, rust, bright red..

Example Model Outputs

GPT-2 (124M)

color	P=0.14
black	P=0.08
rgb	P=0.02

GPT-3 (175B)

navy	P=0.24
dark blue	P=0.13
blue	P=0.08

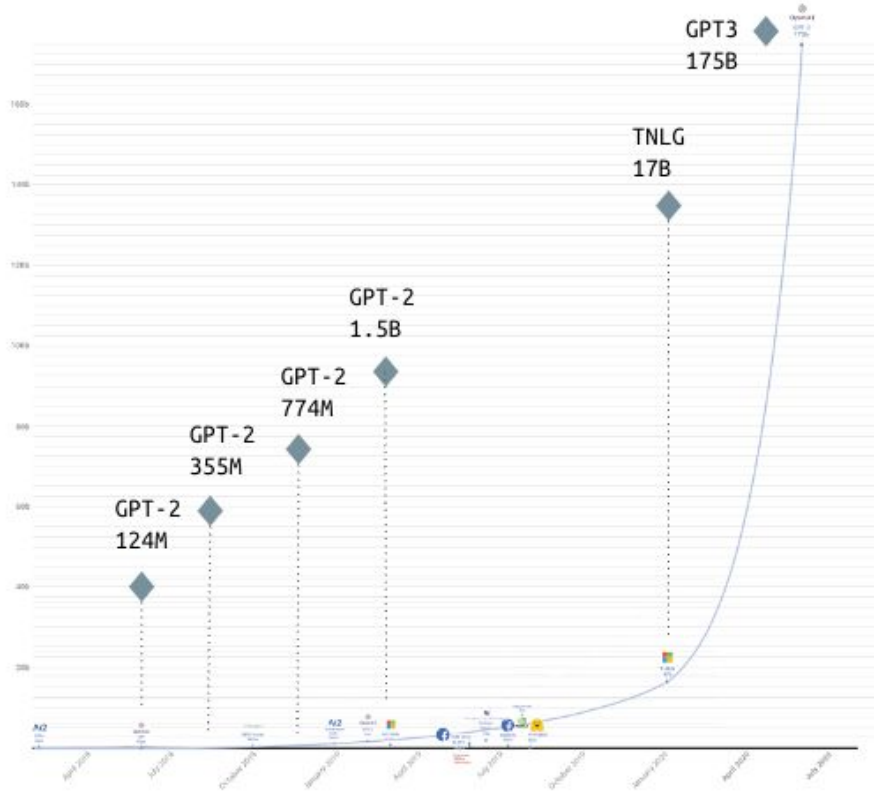
Models tested

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Our models:

GPT-2 (124 M)
GPT-2 (355 M)
GPT-2 (774 M)
GPT-2 (1.5 B)
TNLG (17 B)
GPT-3 (175 B)

All language models trained to generate words sequentially



Few-shot prompting the models

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Prompt style 1

```
Array
[[0. 0. 0. 0. 0. 1.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]
Position:  right

Array
[[0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 1. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0.]]
Position:  right

Array
[[0. 1. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]
Position:  left

Array
[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 1.]
 [0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
Position:  right
```

Prompt style 2

```
Array
[[0. 0. 0. 0. 1. 0. 0.]]
Direction:  right

Array
[[1. 0.]
 [0. 0.]
 [0. 0.]
 [0. 0.]
 [0. 0.]
 [0. 0.]]
Direction:  left

Array
[[0. 1. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0.]]
Direction:  left

Array
[[0. 0. 0. 1. 0.]
 [0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0.]]
Direction:  right
```

Prompt style 3

```
Array
[[0. 0.]
 [0. 1.]
 [0. 0.]
 [0. 0.]
 [0. 0.]
 [0. 0.]
 [0. 0.]]
The direction is:  right

Array
[[0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 1.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0.]]
The direction is:  right

Array
[[0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0.]
 [1. 0. 0. 0. 0.]]
The direction is:  left
```

Example answers

```
left
right
.
.
.
```



How do we ensure models aren't memorising what we're testing?

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How do we ensure models aren't memorising what we're testing?

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Matrix (mathematics)

From Wikipedia, the free encyclopedia

For other uses, see Matrix.
"Matrix theory" redirects here. For the physics topic, see Matrix theory (physics).

In mathematics, a **matrix** (plural **matrices**) is a rectangular array or table of numbers, symbols, or expressions, arranged in rows and columns, which is used to represent a mathematical object or a property of such an object. For example,

$$\begin{bmatrix} 1 & 9 & -13 \\ 20 & 5 & -6 \end{bmatrix}$$

is a matrix with two rows and three columns; one say often a "two by three matrix", a "2×3-matrix", or a matrix of dimension 2×3.

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How do we ensure models aren't memorising what we're testing?

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Description [edit]

A matrix of the form

$$L = \begin{bmatrix} \ell_{1,1} & & & & 0 \\ \ell_{2,1} & \ell_{2,2} & & & \\ \ell_{3,1} & \ell_{3,2} & \ddots & & \\ \vdots & \vdots & \ddots & \ddots & \\ \ell_{n,1} & \ell_{n,2} & \dots & \ell_{n,n-1} & \ell_{n,n} \end{bmatrix}$$

is called a **lower triangular matrix** or **left triangular matrix**, and analogously a matrix of the form

$$U = \begin{bmatrix} u_{1,1} & u_{1,2} & u_{1,3} & \dots & u_{1,n} \\ & u_{2,2} & u_{2,3} & \dots & u_{2,n} \\ & & \ddots & \ddots & \vdots \\ & & & \ddots & u_{n-1,n} \\ 0 & & & & u_{n,n} \end{bmatrix}$$

is called an **upper triangular matrix** or **right triangular matrix**. A lower or left triangular matrix is commonly denoted with the variable *L*, and an upper or right triangular matrix is commonly denoted with the variable *U* or *R*.

Examples [edit]

This matrix

$$\begin{bmatrix} 1 & 4 & 1 \\ 0 & 6 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

is upper triangular and this matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 8 & 0 \\ 4 & 9 & 7 \end{bmatrix}$$

is lower triangular.

How do we ensure models aren't memorising what we're testing?

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	Name	Hex (RGB)	Red (RGB)	Green (RGB)	Blue (RGB)	Hue (HSL/HSV)	Satur. (HSL)	Light (HSL)	Satur. (HSV)	Value (HSV)	CGA number (name); alias
	Aqua	#00FFFF	0%	100%	100%	180°	100%	50%	100%	100%	11 (high cyan); cyan
	Blue	#0000FF	0%	0%	100%	240°	100%	50%	100%	100%	09 (high blue)
	Teal	#008080	0%	50%	50%	180°	100%	25%	100%	50%	03 (low cyan)
	Navy	#000080	0%	0%	50%	240°	100%	25%	100%	50%	01 (low blue)
	Lime	#00FF00	0%	100%	0%	120°	100%	50%	100%	100%	10 (high green); green
	Green	#008000	0%	50%	0%	120°	100%	25%	100%	50%	02 (low green)
	Black	#000000	0%	0%	0%	0°	0%	0%	0%	0%	00 (black)
	Gray	#808080	50%	50%	50%	0°	0%	50%	0%	50%	08 (dark gray)
	Purple	#800080	50%	0%	50%	300°	100%	25%	100%	50%	05 (low magenta)
	Olive	#808000	50%	50%	0%	60°	100%	25%	100%	50%	06 (brown)
	Maroon	#800000	50%	0%	0%	0°	100%	25%	100%	50%	04 (low red)
	Silver	#C0C0C0	75%	75%	75%	0°	0%	75%	0%	75%	07 (light gray)
	White	#FFFFFF	100%	100%	100%	0°	0%	100%	0%	100%	15 (white)
	Fuchsia	#FF00FF	100%	0%	100%	300°	100%	50%	100%	100%	13 (high magenta);

From all the (training) data on the web, there are many instances of the "groundings" we wish to evaluate (e.g., names of colours) that language models may have seen during training.

How do we ensure that we are truly testing grounding and not memorisation?



Isomorphism and rotated worlds

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Isomorphism and rotated worlds

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"..systems are capable of understanding if they emulate representations that are isomorphic to underlying meaning under important semantic relations like equivalence."

$$f(x) = f(x') \iff g(x) = g(x').$$

Can models perform equally well and use the structure/relation between words under different grounding functions for equivalent worlds?

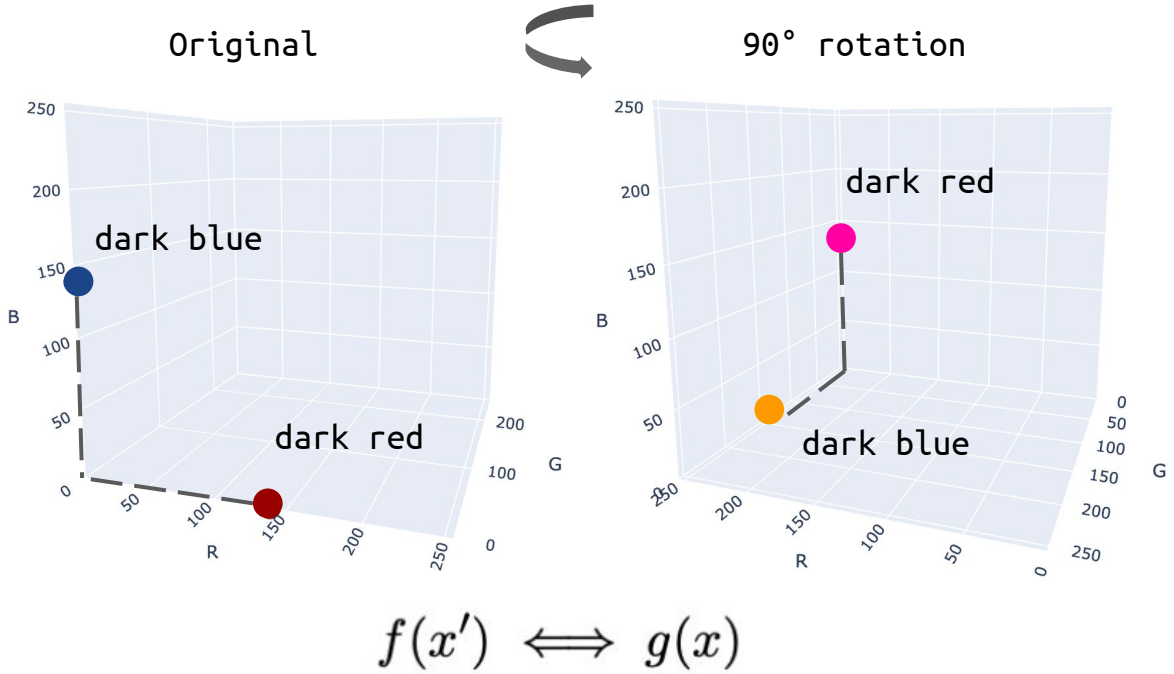
[Merrill et al., 2021]



Isomorphism and rotated worlds

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Example: Rotations in colour space



$$f(x') \iff g(x)$$

New "grounding" or semantic meaning of each element in a rotated world

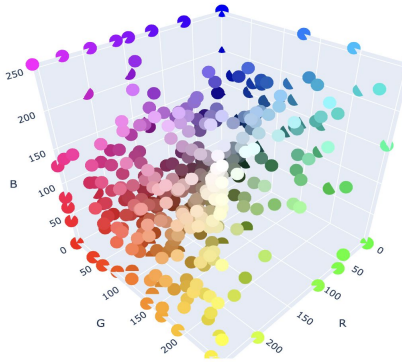
Preserves the structural relation between elements

Isomorphism and rotated worlds

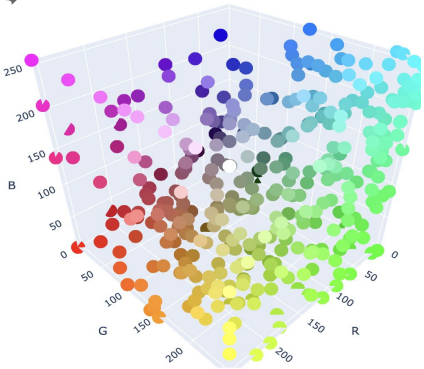
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Rotations in colour space

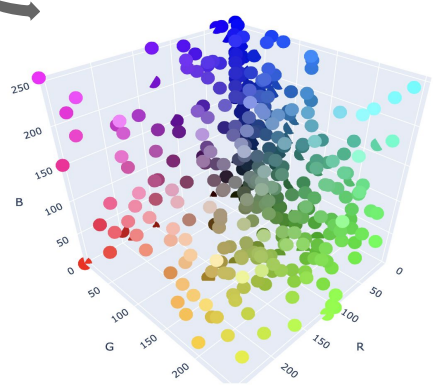
Original



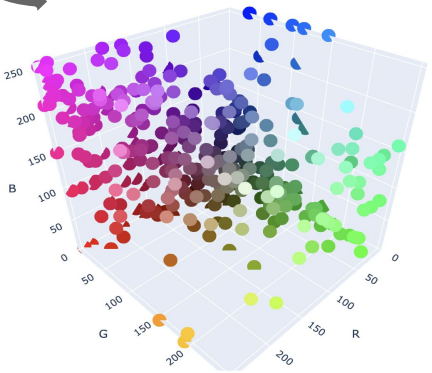
90° rotation



180° rotation



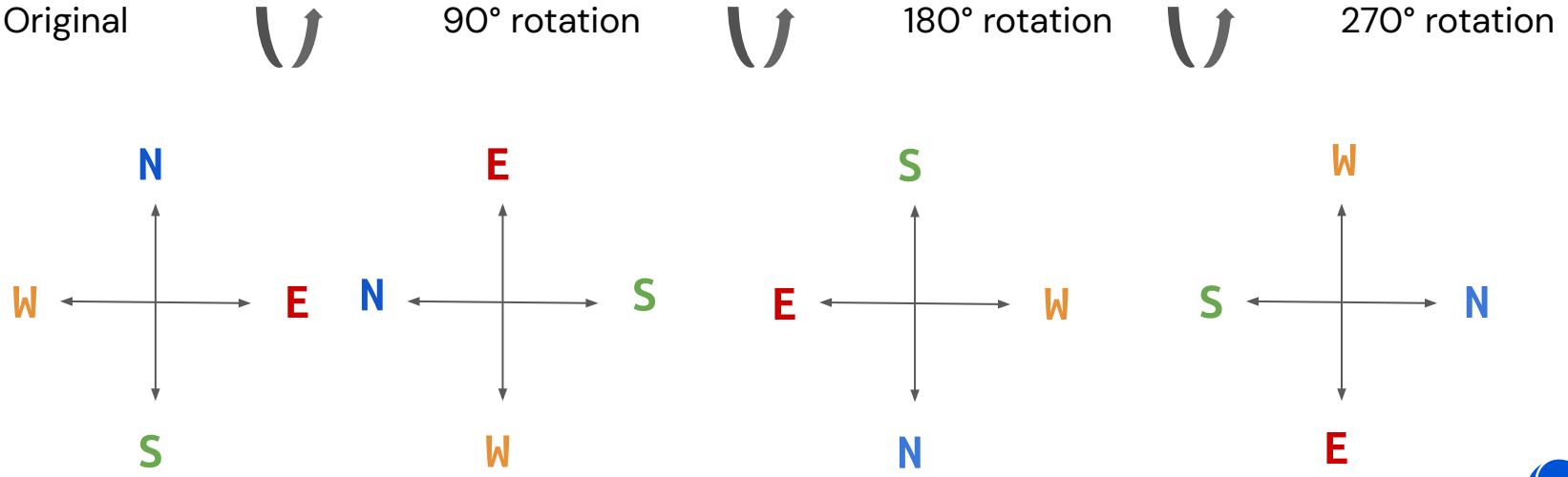
270° rotation



Isomorphism and rotated worlds

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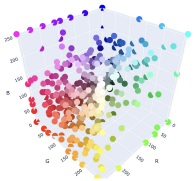
Rotations in cardinal direction space



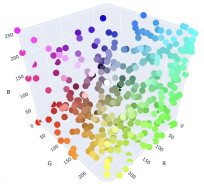
Isomorphism and rotated worlds: our expectation

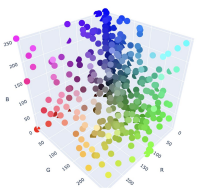
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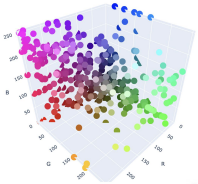
Original



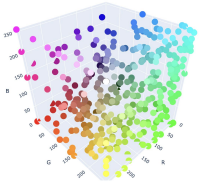
90° or 180° or 270° rotation







Random scramble



$f(x') \iff g(x)$

$f(x') \not\iff g(x)$

Preserves relation between word meanings

Does not preserve relation between word meanings

If models know the relation between words..

And can learn a grounding in one world, they should learn a grounding in all worlds that preserve relations.

The generalisation pattern is somewhat consistent!

A grounding that does not preserve this should be impossible to learn.

The generalisation pattern is random and inconsistent!

Experimental setup

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A sanity check (few-shot generalisation)

Can models learn grounded concepts when it is taught to them during (few-shot) training?

Zero-shot generalisation

If we teach a model some grounded concepts, can they generalise to other related concepts?

Error Analysis

How far are model predictions (in the grounded world) from the true grounding?

Does this hold in the true <-> rotated worlds? Is performance lower in the random worlds?







Few shot generalisation

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Can models learn the meaning of a grounded concept and generalise this meaning to new worlds?



Few shot generalisation

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Can models learn the meaning of a grounded concept and generalise this meaning to new worlds?

Example Input (20 in-context-learning examples followed by prompt)

```
World:
[0. 0. 0.]
[0. 0. 0.]
[0. 0. 1.]
Answer: right

World:
[1. 0. 0. 0.]
Answer: left

...13 more...

World:
[0. 0. 0.]
[0. 0. 0.]
[0. 0. 1.]
Answer: right

World:
[0. 0.]
[1. 0.]
[0. 0.]
Answer: left

World:
[0. 0. 0.]
[0. 0. 0.]
[0. 0. 1.]
[0. 0. 0.]
[0. 0. 0.]
Answer: right

World:
[0. 1. 0. 0.]
Answer: left

World:
[1. 0.]
[0. 0.]
[0. 0.]
[0. 0.]
[0. 0.]
Answer: left

World:
[1. 0. 0. 0.]
[0. 0. 0. 0.]
Answer:
```

Example Model Outputs

GPT-2 (124M)

world	P=0.09
0. 0.]]	P=0.08
[0 [0	P=0.01

GPT-3 (175B)

left	P=0.20
right	P=0.11
leftmost	P=0.01

Few shot generalisation

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Top-3 Accuracy

Top-1 Accuracy

In new unseen worlds, models ***can*** learn to ground what is up vs. down when they have been taught (few-shot) both concepts.

```
Array
[[0. 0. 0. 1.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]]
The direction is:  up
```

Zero shot generalisation

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Can models learn the meaning of some grounded concepts and apply this to learn the meaning of unseen concepts through their relational structure?



Zero shot generalisation

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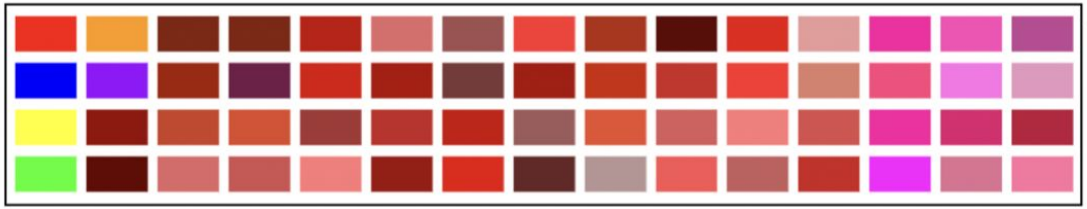
Example Input (60 in-context-learning examples followed by prompt)

RGB: (48, 213, 200)
Answer: orange

RGB: (220, 20, 60)
Answer: crimson

RGB: (0, 0, 128)
Answer:

All 60 Training Examples



6 Primary and Secondary Colours

red, blue, yellow, green, orange, violet

57 Colours Within a Sub-space

dark red, maroon, crimson, fuchsia, rust, bright red..

Example Model Outputs

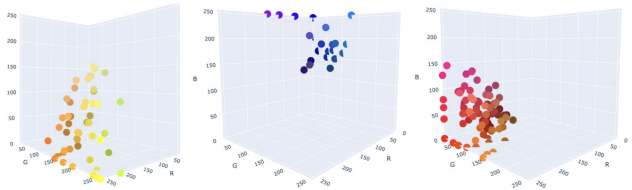
GPT-2 (124M)

color	P=0.14
black	P=0.08
rgb	P=0.02

GPT-3 (175B)

navy	P=0.24
dark blue	P=0.13
blue	P=0.08

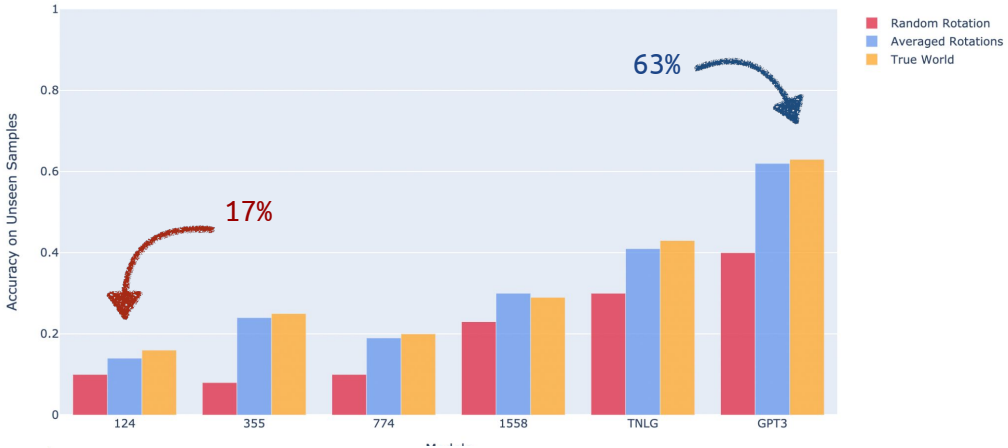
Sub-space centered around the colour red



Example colour sub-spaces for yellow, blue and red

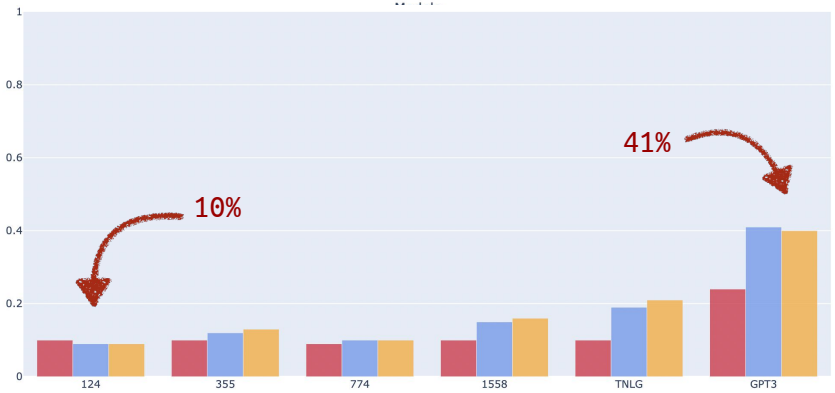
Zero shot generalisation

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Top-3 Accuracy

In new unseen worlds, models have to ground what *unseen colours* when they have been taught only *10 colours* (zero-shot).



Top-1 Accuracy

```
RGB: (0, 65, 106)
Answer: dark imperial blue

RGB: (111, 0, 255)
Answer: electric indigo

RGB: (239, 222, 205)
Answer:
```

Zero shot generalisation

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	Top-1 Accuracy						Top-3 Accuracy					
	Spatial			Cardinal			Spatial			Cardinal		
	Orig.	Rot.	Rand.	Orig.	Rot.	Rand.	Orig.	Rot.	Rand.	Orig.	Rot.	Rand.
R-IV	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
R-ID	0.16	0.16	0.16	0.13	0.13	0.13	0.16	0.16	0.16	0.13	0.13	0.13
124 M	0.11	0.10	0.10	0.13	0.12	0.11	0.23	0.21	0.10	0.25	0.24	0.10
355 M	0.12	0.12	0.10	0.11	0.14	0.10	0.24	0.25	0.15	0.23	0.14	0.12
774 M	0.08	0.09	0.10	0.11	0.12	0.11	0.12	0.19	0.14	0.18	0.17	0.11
1.7 B	0.10	0.11	0.11	0.10	0.11	0.10	0.11	0.18	0.15	0.12	0.12	0.13
175 B	**0.45**	**0.44**	**0.16**	**0.43**	**0.46**	**0.18**	**0.76**	**0.75**	**0.19**	**0.88**	**0.76**	**0.21**

Table 1: Table shows evaluations for generalisation to unseen worlds. Rows show model sizes for GPT-2 models (124M to 1.5B) and the 175B GPT-3 model. Columns show metrics for the original, rotated and random worlds for each of the concept categories. We report both Top-1 and Top-3 accuracy in the first and second columns of each rotated world.



Analysing model predictions

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Analysing model predictions

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Model Prediction	True Grounding
dark red	wine
medium apricot	apricot
burnt orange	orange
green	forest green
aquamarine	sea green
dark rose	deep carmine
light red	saffron
black	ebony
light purple	lavender
.	.
.	.
.	.

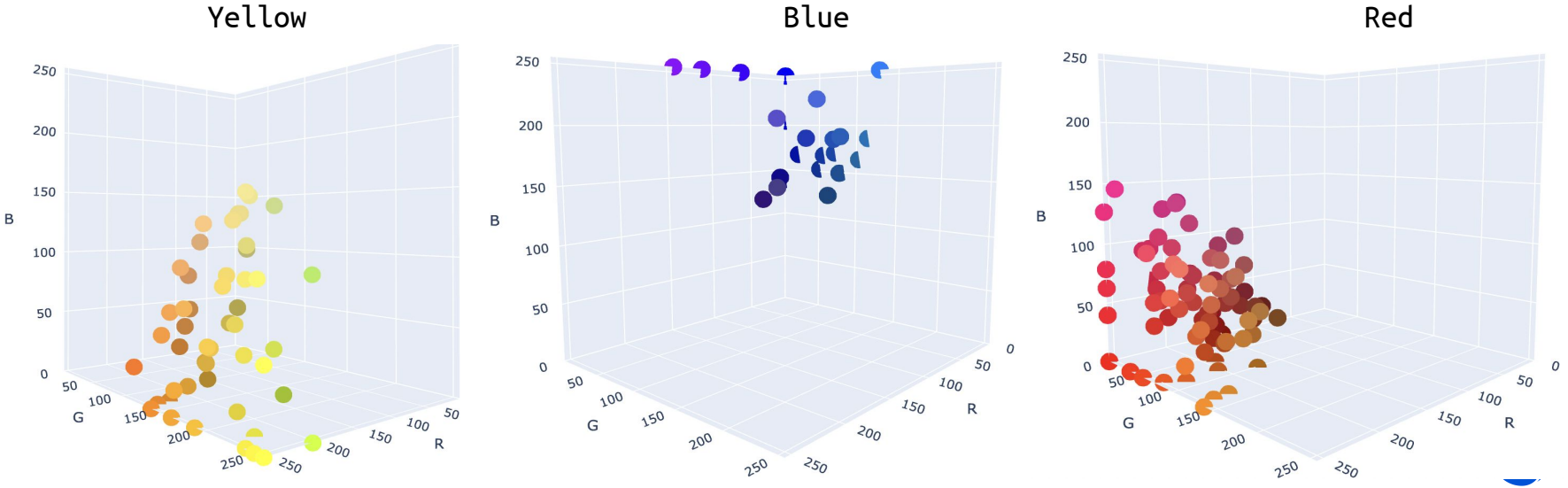
GPT3



Analysing model predictions

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Measure Euclidean distance b/w model predictions and true answer

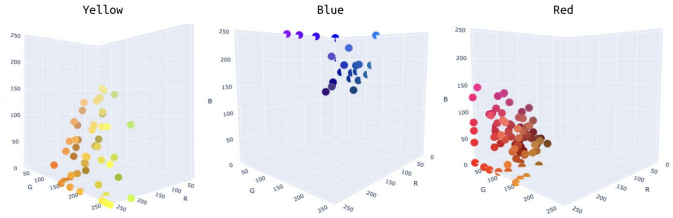


Analysing model predictions

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$$\mathrm{d}(c_1, c_2) = \begin{cases} \sqrt{(c_{1x} - c_{2x})^2 + (c_{1y} - c_{2y})^2 + (c_{1z} - c_{2z})^2} & \text{for} \quad c_1 \in C \\ 500 & \text{for} \quad c_1 \notin C \end{cases}$$

Eq 1. Euclidean distance between two points in space



Analysing model predictions

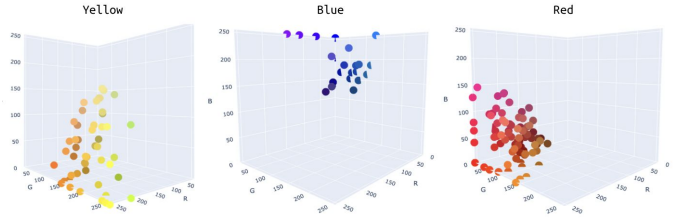
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	124M	355M	774M	1.5B	175B
C	328.3	309.5	209.6	190.7	96.3
R-IV	334.9	334.9	334.9	334.9	334.9
R-ID	174.9	174.9	174.9	174.9	174.9

Table 1. Average Euclidean distance between predictions and ground-truth answers.

$$\mathrm{d}(c_1, c_2) = \begin{cases} \sqrt{(c_{1x} - c_{2x})^2 + (c_{1y} - c_{2y})^2 + (c_{1z} - c_{2z})^2} & \text{for} \quad c_1 \in C \\ 500 & \text{for} \quad c_1 \notin C \end{cases}$$

Eq 1. Euclidean distance between two points in space



Analysing model predictions

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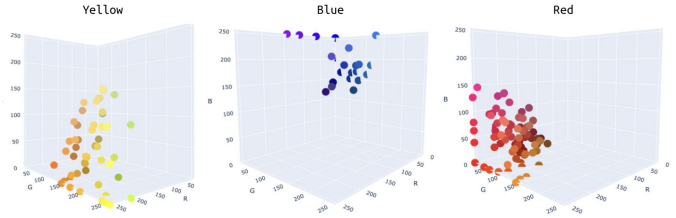
Table 1. Average Euclidean distance between predictions and ground-truth answers.

True G	Predicted G & Distance
dark red	wine (76.5), light crimson (208.1) dark slate gray (144.7)
light green	beige (126.6), light sea green (129.7) cerulean (185.7), violet (262.6)

Table 2. Example predictions from GPT-3 with Euclidean distances from the ground-truth.

$$d(c_1, c_2) = \begin{cases} \sqrt{(c_{1x} - c_{2x})^2 + (c_{1y} - c_{2y})^2 + (c_{1z} - c_{2z})^2} & \text{for } c_1 \in C \\ 500 & \text{for } c_1 \notin C \end{cases}$$

Eq 1. Euclidean distance between two points in space



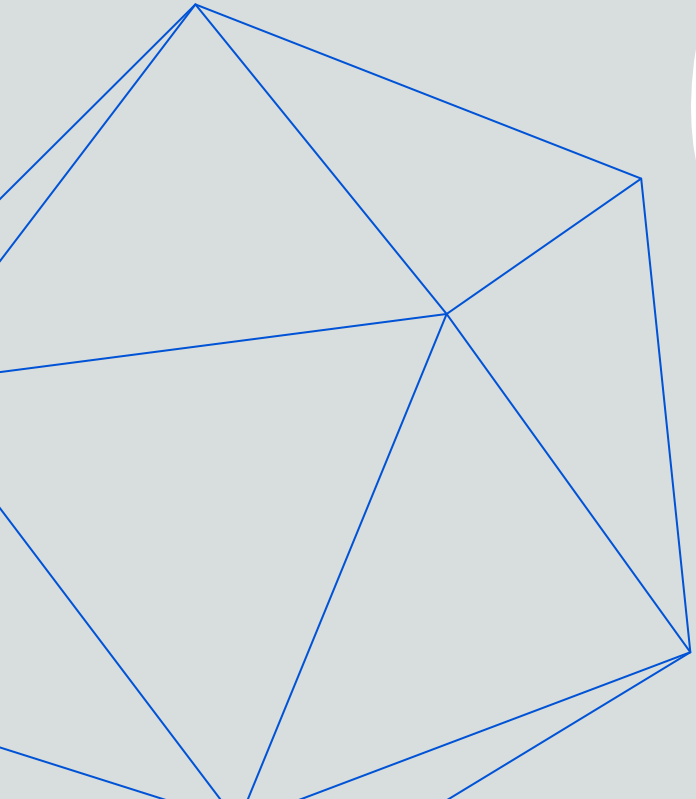
In summary

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1. There exists some amount of meaningful structure in the conceptual spaces formed by language models
2. We can only test this in a small set of domains that can be represented in text
3. Is this useful if what we want is to be able to ground/transfer visual/sensory information to models?



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Paper # 2

Linearly Mapping from Image to Text Space

Mapping between conceptual spaces of different pre-trained models

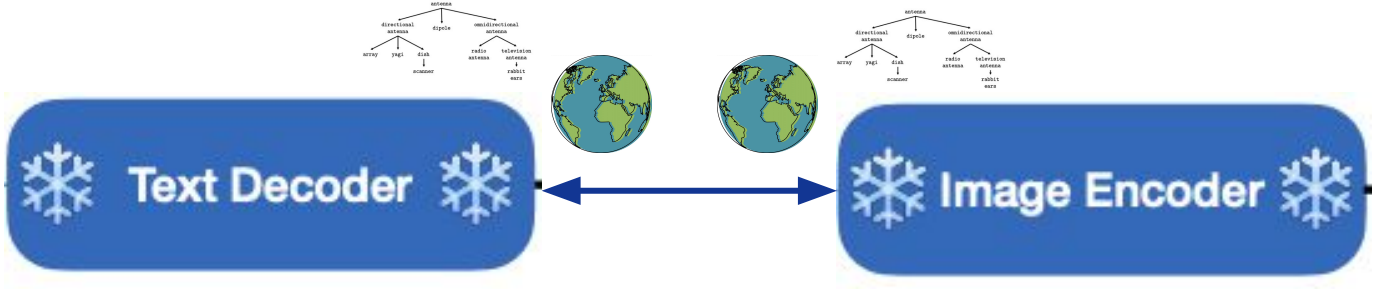
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[Merullo et al., 2022]

Mapping between conceptual spaces of different pre-trained models

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If models do seem to learn grounded representations consistent with the world...



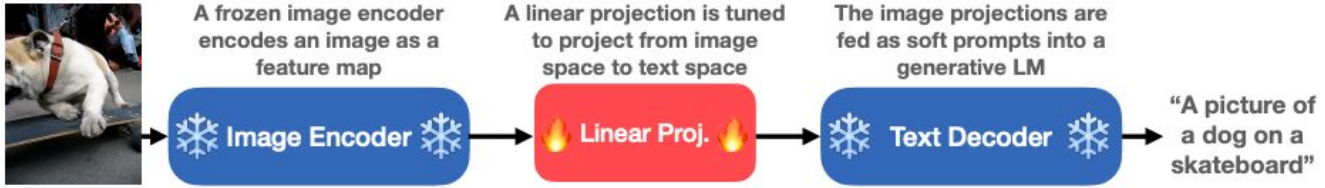
Representations learned by a large LLM e.g., Chinchilla

Representations learned by a large VLM/VM e.g., Flamingo

Can we map language model spaces to visual groundings?

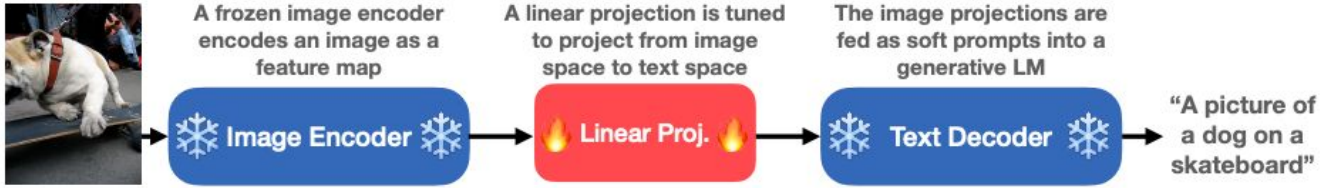
Mapping between conceptual spaces of different pre-trained models

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Mapping between conceptual spaces of different pre-trained models

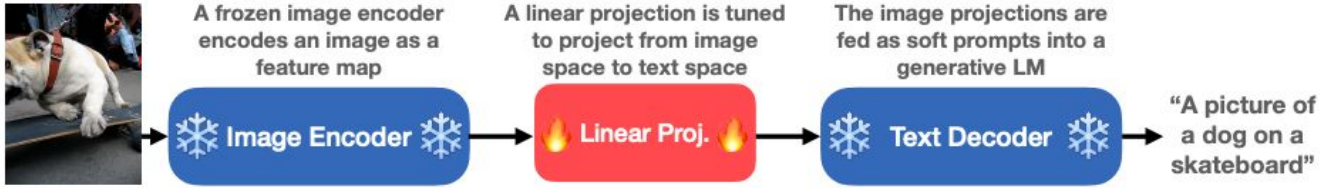
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Question. Can we project image representations from a VLM into the language representation space of an LLM (keeping the rest of the LLM frozen)?

Mapping between conceptual spaces of different pre-trained models

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Question. Can we project image representations from a VLM into the language representation space of an LLM (keeping the rest of the LLM frozen)?

How? Learn a linear projection from image representations created by the VLM into embedding space of the LLM and such that the LLM outputs match true captions on an image captioning task

Mapping between conceptual spaces of different pre-trained models

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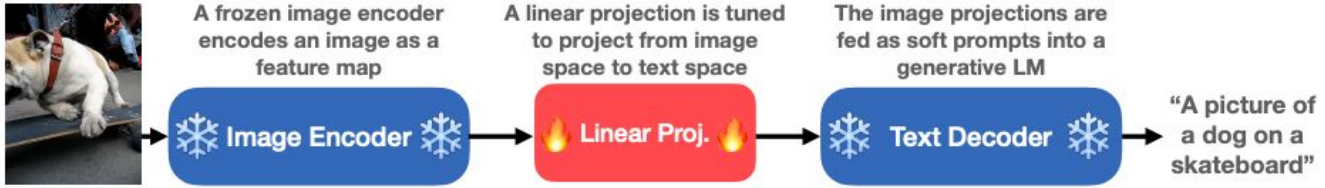
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Why? Since a linear project is distance-preserving, if the conceptual spaces of models are structured similarly, this mapping should be learned (LLM produces correct language for images)

Mapping between conceptual spaces of different pre-trained models

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Question. Can we project image representations from a VLM into the language representation space of an LLM (keeping the rest of the LLM frozen)?

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Why? Since a linear project is distance-preserving, if the conceptual spaces of models are structured similarly, this mapping should be learned (LLM produces correct language for images)

Test. Evaluate how well these projections between two different models can accurately reflect concepts in the image

Evaluating differently trained image encoders

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Evaluating differently trained image encoders

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Evaluating differently trained image encoders

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1. Three different encoders that received different amounts of linguistic supervision during training
    a. NFRN50 was pre-trained on image classification over 1k categories
    b. BEIT had a masked prediction objective with no exposure to language/classes
    c. CLIP was pre-trained to align images with full natural language captions
    d. Control: randomly initialised NFRN50 that was not pretrained
2. Text decoder: GPT-J (6B parameters)

Result #1: Linguistic supervision affects performance

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Result #1: Linguistic supervision affects performance

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1. Linguistic supervision during pre-training plays a key role in concept formation
2. CLIP and NFRN50 outperform BEIT in terms of transferring image properties to language
    a. BEIT only transfers coarse-grained visual properties (scene/background)
    b. CLIP can reproduce concepts in the image to a greater accuracy



Result #1: Linguistic supervision affects performance

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Projection-based captioning is far above a random baseline, and more linguistic information during pre-training allows the projections to be learnt better

Image Captioning	NoCaps - CIDEr-D				NoCaps (All)		CoCo	CoCo	
	In	Out	Near	All	CLIP-S	Ref-S	CIDEr-D	CLIP-S	Ref-S
🔥NFRN50 Tuned	20.9	30.8	25.3	27.3	66.5	72.5	35.3	69.7	74.8
🔥MAGMA (released)	18.0	12.7	18.4	16.9	63.2	68.8	**52.1**	76.7	79.4
🔥MAGMA (ours)	**30.4**	**43.4**	**36.7**	**38.7**	74.3	78.7	47.5	75.3	**79.6**
❄BEIT Random	5.5	3.6	4.1	4.4	46.8	55.1	5.2	48.8	56.2
❄NFRN50 Random	5.4	4.0	4.9	5.0	47.5	55.7	4.8	49.5	57.1
❄BEIT	20.3	16.3	18.9	18.9	62.0	69.1	22.3	63.6	70.0
❄NFRN50	21.3	31.2	26.9	28.5	65.6	71.8	36.2	68.9	74.1
❄BEIT FT.	**38.5**	**48.8**	**43.1**	**45.3**	73.0	78.1	51.0	74.2	78.9
❄CLIP	34.3	48.4	41.6	43.9	**74.7**	**79.4**	**54.9**	**76.2**	**80.4**



Result #2: Evaluating language generations

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Result #2: Evaluating language generations

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Image Captioning



CLIP	a giraffe in the lobby of the building
NFRN50	the giraffe in the zoo.
BEIT	a peacock in the garden
NFRN50 Random	a man and a woman in a field of flowers



CLIP	tennis player in action
NFRN50	tennis player at the tennis tournament.
BEIT	tennis player during a tennis match.
NFRN50 Random	the new logo for the team

Visual Question Answering



CLIP	He is surfing a wave.
NFRN50	He is surfing the waves.
BEIT	He is jumping into the water.
NFRN50 Random	He is swimming in the pool.

Q: What is the person doing?
A: `surfing`

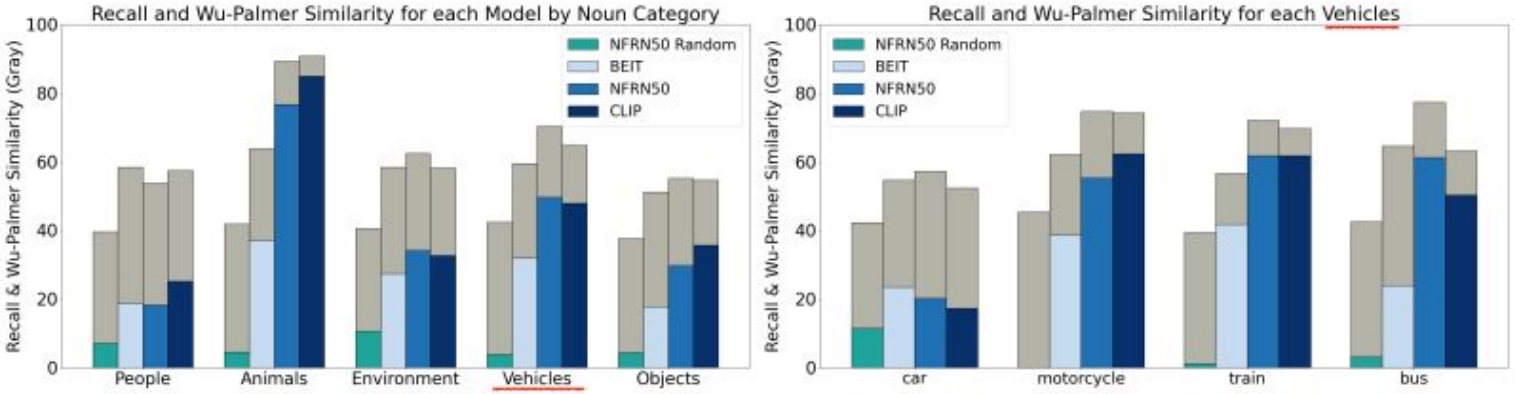


CLIP	A tennis racket
NFRN50	A tennis racket
BEIT	A baseball bat.
NFRN50 Random	A tree

Q: What is the person holding?
A: `tennis racket`

Result #2: Evaluating language generations

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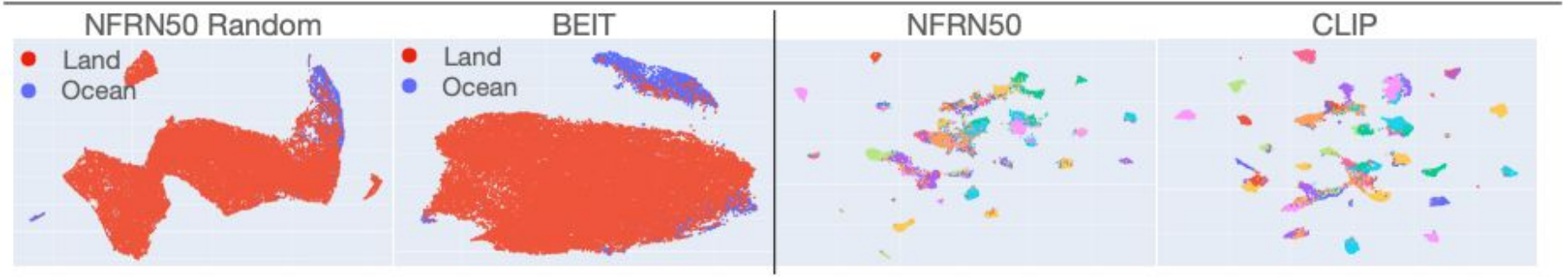


Models with more linguistic supervision have better performance (recall) of nouns and Wu-Palmer similarity (predicting nouns close in meaning based on taxonomy/tree relations).

Results show that some models (BEIT) have low recall for exact-word prediction, they predict related visually similar concepts (e.g., saying “car” instead of “bus”)

Evaluating image representations

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(b) **UMAP projections of AWA images:** While NFRN50 and CLIP cluster tightly along lexical categories (color coded by animal), BEIT clusters the most distinctly along animals that live in water/the ocean; the randomly initialized NFRN50 mostly randomly overlap in one cluster.

Image representation clusterings seem to reflect the language generation results.
NFRN50 and CLIP representations cluster tightly along lexical categories, but BEIT tends to cluster most strongly by perceptual features (e.g., habitat/having flippers).

In summary

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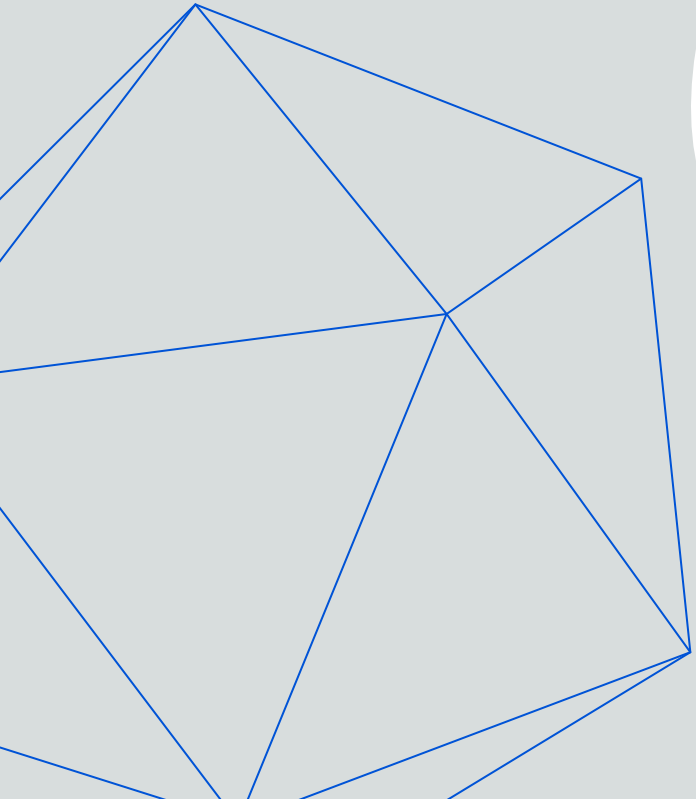
1. **Paper # 1.** Models that have trained on text-only data learn a conceptual space that allows them to ground concepts in textual worlds

2. **Paper # 2.** Models that have trained on text-only data can be mapped to models that train on image data (and both models are trained separately!) and can transfer conceptual information from one another



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Failures of LMS & VLMS



LLMs have never been grounded during training

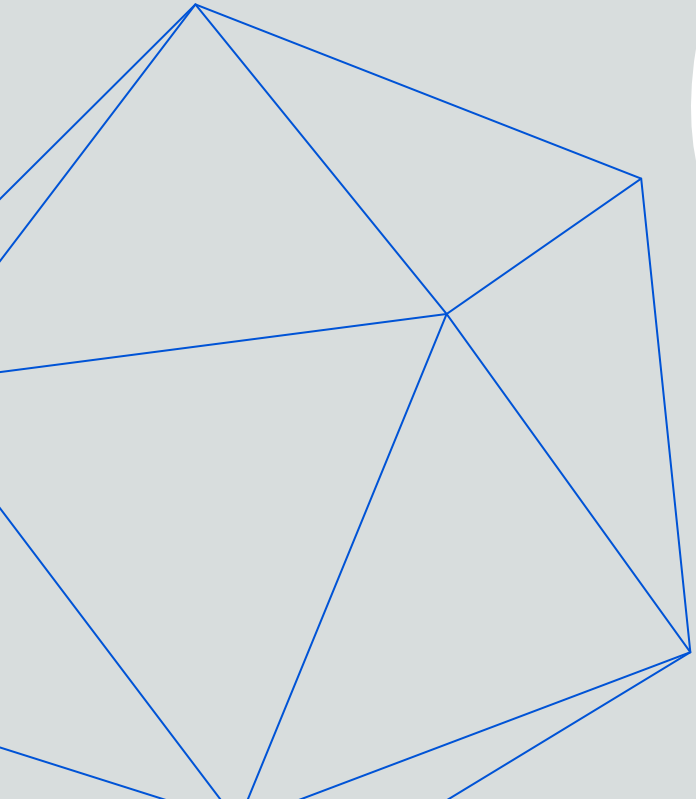
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Fact. Stark contrast to the way humans process language and reason about concepts

Question? Do the visual interpretations of concepts that VLMs represent resemble how humans reason about those concepts?



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Paper # 3

DALLE-2 is Seeing Double: Flaws in Word-to-Concept Mapping in Text2Image Models

How do word roles map to concepts for VLMs?

Evaluate VLMs on linguistic stimuli that conflate concept role/meaning.

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How do word roles map to concepts for VLMs?

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Figure 1: The word *bat* is realized as two entities given the prompt *a bat is flying over a baseball stadium*.

How do word roles map to concepts for VLMs?

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Figure 1: The word *bat* is realized as two entities given the prompt *a bat is flying over a baseball stadium.*

1. A word or phrase is interpreted as two distinct (concurrently-incompatible entities)
   a. *“a bat is flying over a stadium”*

How do word roles map to concepts for VLMs?

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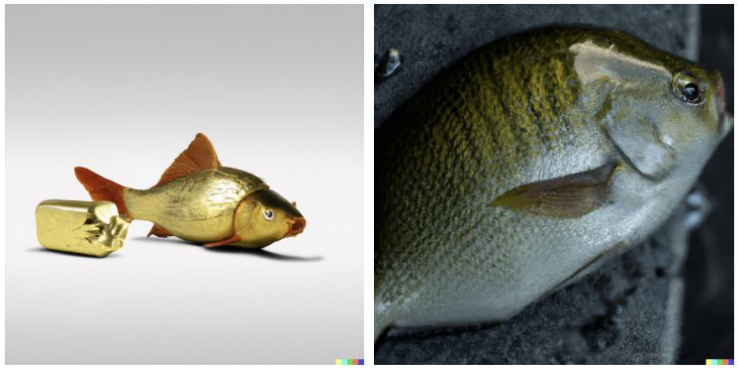
Figure 1: The word *bat* is realized as two entities given the prompt *a bat is flying over a baseball stadium*.

1. A word or phrase is interpreted as two distinct (concurrently-incompatible entities)
   a. *"a bat is flying over a stadium"*

80.3% over 216 images (17 stimuli)

How do word roles map to concepts for VLMs?

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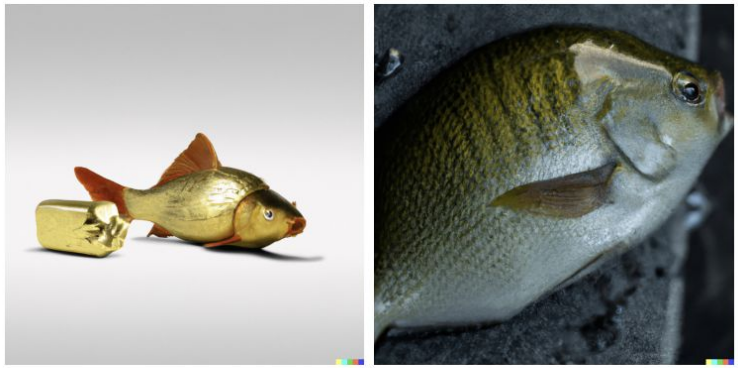
Main (left): *a **fish** and a **gold ingot**.*
Control (right): *a fish and an ingot.*

1. A word or phrase is interpreted as two distinct (concurrently-incompatible entities)
   a. *"a bat is flying over a stadium"*

2. A word is interpreted as a modifier of two different entities
   a. *"a fish and a gold ingot"*

How do word roles map to concepts for VLMs?

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Main (left): *a **fish** and a **gold ingot**.*
Control (right): *a fish and an ingot.*

1. A word or phrase is interpreted as two distinct (concurrently-incompatible entities)
   a. *"a bat is flying over a stadium"*

2. A word is interpreted as a modifier of two different entities
   a. *"a fish and a gold ingot"*

97.2% over 6 stimuli-control pairs

How do word roles map to concepts for VLMs?

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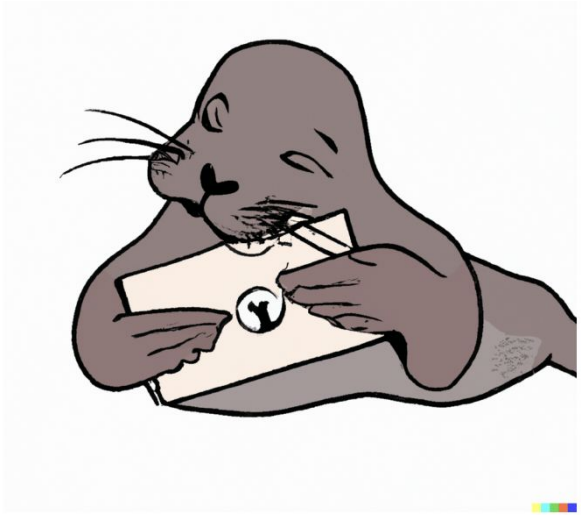


Figure 2: The word *seal* is realized both as an object and as a modifier of another word (*letter*) in the prompt *a seal is opening a letter*, resulting in *concept leakage*. The prompt tends to generate *sealed* letters, while minimally-different prompts that do not contain the noun *seal* do not.

1. A word or phrase is interpreted as two distinct (concurrently-incompatible entities)
   a. *"a bat is flying over a stadium"*

2. A word is interpreted as a modifier of two different entities
   a. *"a fish and a gold ingot"*

3. A word is interpreted simultaneously as an entity and as a modifier of a different entity
   a. *"a seal is opening a letter"*

How do word roles map to concepts for VLMs?

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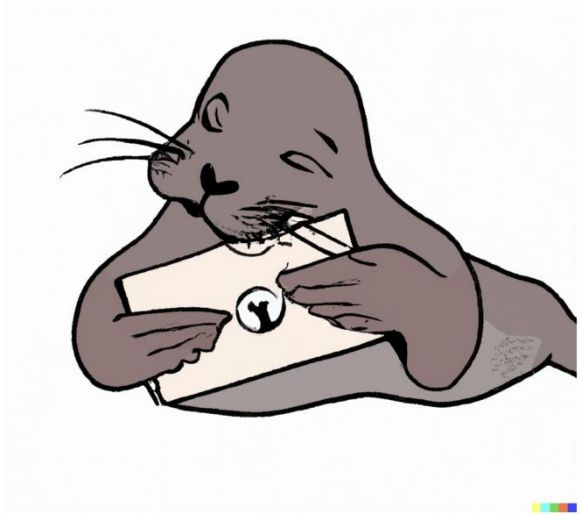


Figure 2: The word *seal* is realized both as an object and as a modifier of another word (*letter*) in the prompt *a seal is opening a letter*, resulting in *concept leakage*. The prompt tends to generate *sealed* letters, while minimally-different prompts that do not contain the noun *seal* do not.

1. A word or phrase is interpreted as two distinct (concurrently-incompatible entities)
   a. *"a bat is flying over a stadium"*

2. A word is interpreted as a modifier of two different entities
   a. *"a fish and a gold ingot"*

3. A word is interpreted simultaneously as an entity and as a modifier of a different entity
   a. *"a seal is opening a letter"*

92.5% over 10 stimuli-control pairs

Takeaways

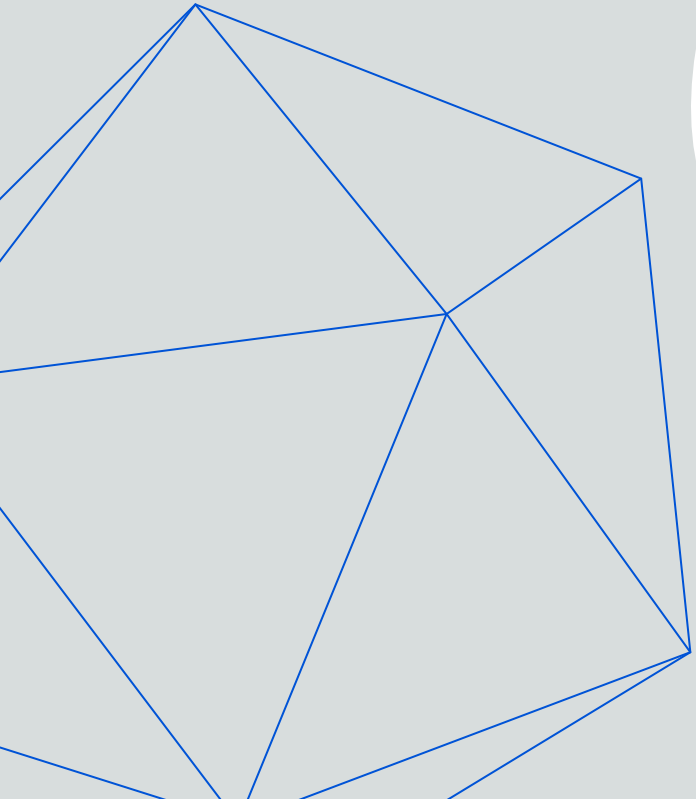
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1. Caveats: small-scale experiments, need a more comprehensive evaluation
2. VLMs don't follow the **basic principles of symbol-to-entity mapping**
3. A single referent in a sentence is not mapped to one grounding in the world but **concept leakage** occurs
4. Leakage reflects some differences in the conceptual spaces that the models form vs. what humans form



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The future?



LLMS / VLMs have shown remarkably good performance

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ChatGPT

DALL-E

Zero/few shot

Can generate fluent, coherent and human-like text (QA, summarisation)

Title: United Methodists Agree to Historic Split Subtitle: Those who oppose gay marriage will form their own denomination Article: **After two days of intense debate, the United Methodist Church has agreed to a historic split - one that is expected to end in the creation of a new denomination, one that will be "theologically and socially conservative," according to The Washington Post. The majority of delegates attending the church's annual General Conference in May voted to strengthen a ban on the ordination of LGBTQ clergy and to write new rules that will "discipline" clergy who officiate at same-sex weddings.**

Language Models are Few-Shot Learners. Brown et. al., NeuRIPs 2020.

Can few-shot learn NLU tasks: entailment, sentiment, grammatical corrections

Poor English input: The patient was died.
Good English output: **The patient died.**

Poor English input: We think that Leslie likes ourselves.
Good English output: **We think that Leslie likes us.**

Poor English input: Janet broke Bill on the finger.
Good English output: **Janet broke Bill's finger.**

Poor English input: Mary arranged for, in St. Louis, John to rent a house cheap.
Good English output: **Mary arranged for John to rent a house in St. Louis.**

Context →	The bet, which won him dinner for four, was regarding the existence and mass of the top quark, an elementary particle discovered in 1995.
question:	The Top Quark is the last of six flavors of quarks predicted by the standard model theory of particle physics. True or False?
answer:	
Target Completion →	False

Figure G.31: Formatted dataset example for RTE

GLUE: A Multi-task Benchmark and Analysis Platform for Natural Language Understanding. Wang et. al., ICLR 2019.

Can reconstruct WordNet structure and learn all word relations!

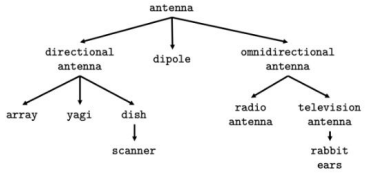
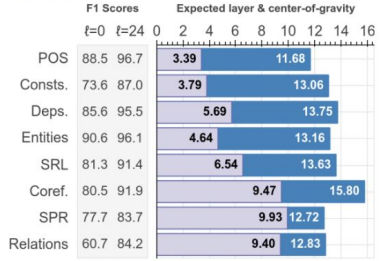


Figure 1: *An example subtree from the* WordNet *hierarchy.*

Constructing Taxonomies from Pretrained Language Models. Chen, Lin and Klein. NAACL 2021.

Can solve most classical NLP tasks!



BERT Rediscovers the Classical NLP Pipeline. Tenney, Das and Pavlick. ACL 2019.

They have some semblance of a conceptual structure that we think is reasonable

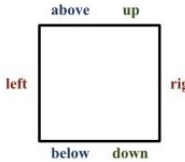
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ChatGPT

DALL-E

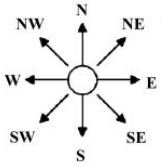
Zero/few shot

Spatial Terms



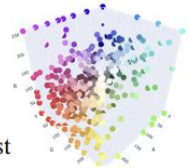
World:
[0. 0. 0. 0. 0.]
[0. 0. 0. **1**. 0.]
[0. 0. 0. 0. 0.]
Answer: right

Cardinal Directions



World:
[0. 0. **1**.]
[0. 0. 0.]
[0. 0. 0.]
Answer: northeast

RGB Colours



RGB: (255, 0 0)
Answer: red

RGB: (143, 0, 255)
Answer: violet



A frozen image encoder encodes an image as a feature map

Image Encoder

A linear projection is tuned to project from image space to text space

Linear Proj.

The image projections are fed as soft prompts into a generative LM

Text Decoder

"A picture of a dog on a skateboard"

Fundamental flaws in text-only supervised pre-training

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ChatGPT

DALL-E

→ Zero/few shot



1. Ungrounded, therefore meaning only exists in textual symbol space
2. Limits to what can be learned: no access to sensory information that humans receive (*heat, touch..*)
3. Limits to what can be generated: restricted textual output spaces
4. Makes it hard to transfer to grounded/immersive environments we care about

Fundamental flaws in text-only supervised pre-training

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If there do seem to internal inconsistencies/missing information, we might not want to use ungrounded / frozen models in zero/few-shot ways

Question? Can we keep pre-trained layers but update parameters to reflect grounded conceptual structure

Keep pre-trained layers but update parameters to reflect grounded conceptual structure

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Post-hoc fine-tuning with

a. Reinforcement learning instead of supervised learning
   i. *alignment with intent, maybe conceptual structure*

b. Non-textual data instead of text-only data
   i. *learn grounded meaning through action, vision*

c. Multiple agents instead of one agent trained on data
   i. *reason about beliefs, agency, pragmatics, communicative intent*

Evaluating representations for conceptual structure

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"What concepts have to be" (Fodor, 1998)

Evaluate whether neural network representations encode and understand concepts analogous to human understanding of concepts.

a. Concepts are represented modularly
b. Concepts are causal and represent types

Unit Testing for Concepts in Neural Networks [Lovering and Pavlick 2022]

Mechanistic Interpretability

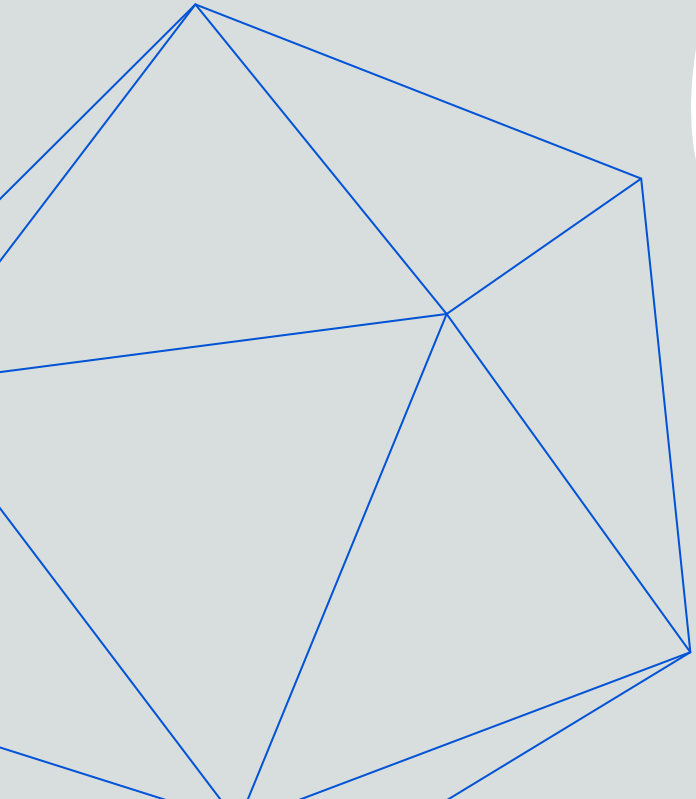
Evaluate whether neural network components (circuits, paths of activations) represent these concepts

a. Understand at an algorithmic level what attention heads correspond to
b. Causally intervene on model components to influence behaviour

Interpretability in the Wild [Wang et. al., 2022]



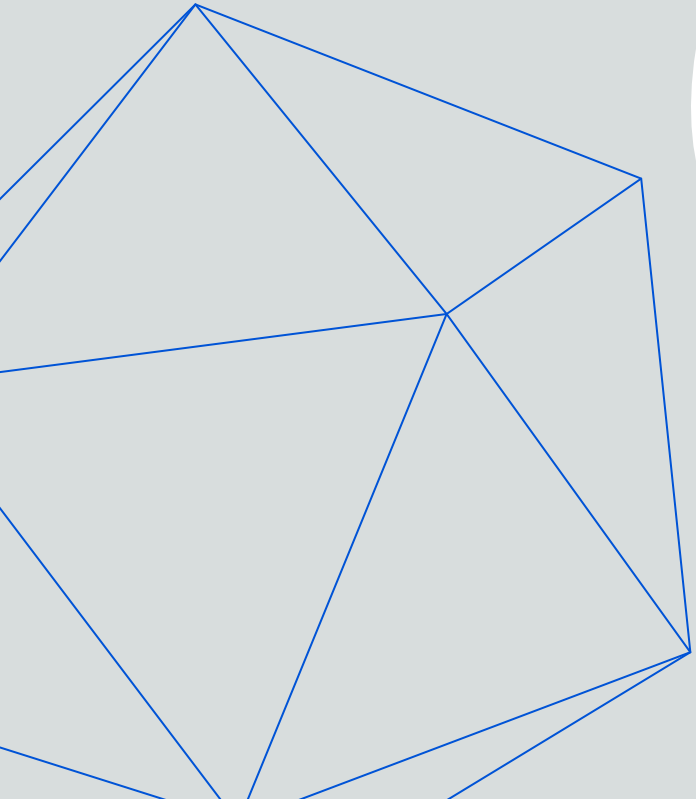
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Thank you

Questions?

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Other related work

Pretrained LMs that transfer to the world

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Pretrained Transformers As Universal Computation Engines

Kevin Lu
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Facebook AI Research
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Multimodal Few-Shot Learning with Frozen Language Models

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Felix Hill
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Link

MAGMA – Multimodal Augmentation of Generative Models through Adapter-based Finetuning

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Link



Comprehensive evaluation of concepts in agents

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Evaluation Beyond Task Performance: Analyzing Concepts in AlphaZero in Hex

Link

Charles Lovering* **Jessica Zosa Forde***
George Konidaris **Ellie Pavlick** **Michael L. Littman**
Department of Computer Science
Brown University
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Unit Testing for Concepts in Neural Networks

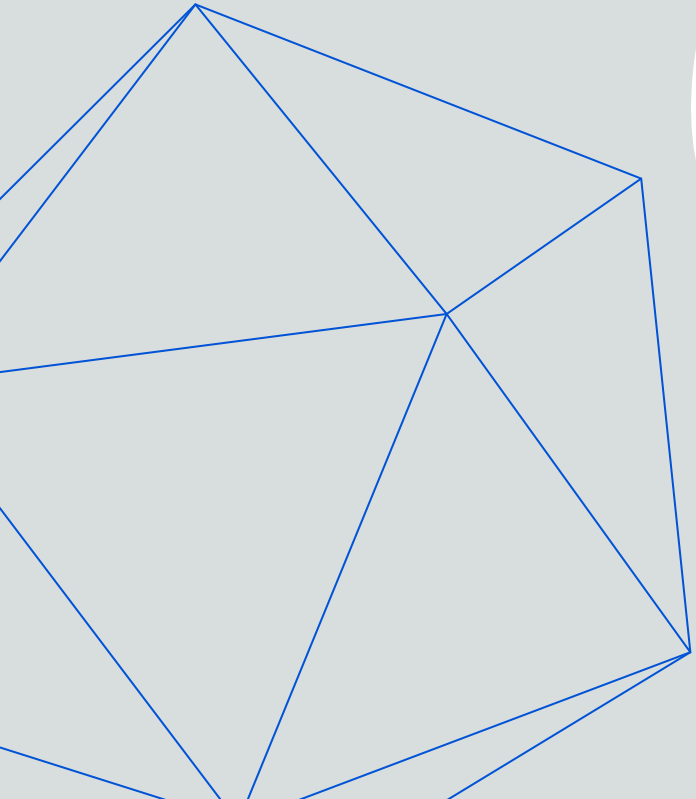
Link

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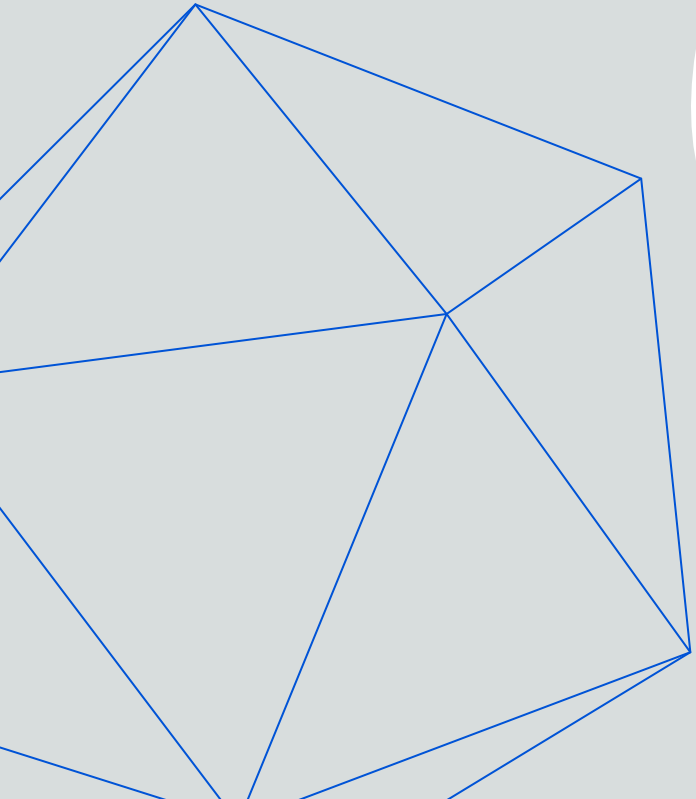
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Thank you

Questions?

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Extra slides



Compositional concepts

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```
RGB: (255, 165, 0)
Answer: orange

RGB: (255, 179, 71)
Answer: pastel orange

RGB: (255, 192, 203)
Answer: pink

RGB: (222, 165, 164)
Answer: pastel pink

RGB: (255, 0, 0)
Answer: red

RGB: (255, 105, 97)
Answer: pastel red

RGB: (143, 0, 255)
Answer: violet

RGB: (203, 153, 201)
Answer: pastel violet
```

Test on blue / pastel blue..



Continuous prompts

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Continuous prompts

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1. Instead of just hyper-parameter searching over prompt space for best task performance ***learn*** which prompts help task performance

2. For each task, freeze the LLM but learn *k* embeddings (attached to embedding layer of LLM) while training for task performance

3. At test time, this projection tells you which tokens to prepend to the prompt

4. This reliably improves performance on downstream tasks

5. Worth noting terminology over papers: “prefix-tuning”, “soft prompts”, “continuous prompts” that all refer to the process of freezing the LLM but training some sub-component that can be added at test-time



Continuous prompts from images

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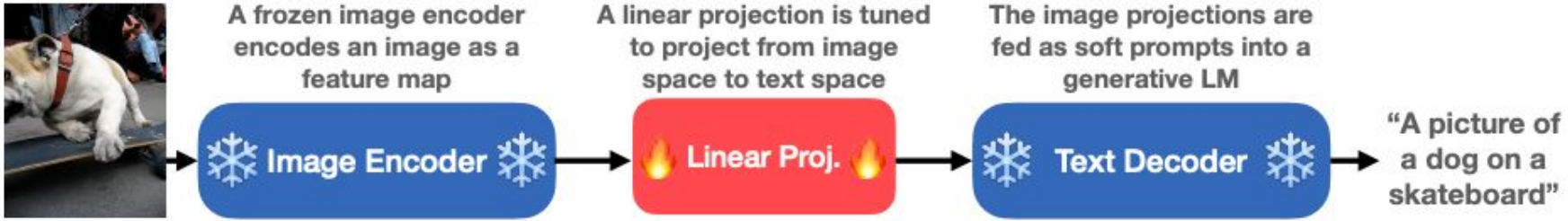


Figure 1: We train linear projections from image representations into the input space of a language model to produce captions describing images. We find that LMs can describe the contents of most image representations, but performance varies based on the type of image encoder used.

Analysing model predictions

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Example related generations:

```
magenta
green
aqua
red
orange
purple
yellow
blue
dark blue
tan
silver
cyan
.
.
.
```

124M

Example unrelated generations:

```
depending on your key length (48 bits, no 32 bits),
you'll get a variety of brightness values: red = 0,
green = 6.5, blue = 0.5 and orange = 5 pixels (4 bytes
on ordered hexadecimal).

zeroes

the lossy variety existed a few blemishes in the hood,
irons, axles etc and the tough new moptoid area
correctibility repairbox with drivers and display was
marked as being in good repair. this recall later
proved to be property of the underhanded mismanaging
field operations center on the back more than 50 years
later when…

cape high
```



Analysing model predictions

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Example related generations:

deep magenta
mint
dark chocolate
light purple
light bordeaux
tan
silver
cyan
charcoal
green
peach
deep indigo
.
.
.

175B

Example unrelated generations:

liver
sherbet
rhodolite

Almost no unrelated generations!



Training language agents together

1. Formulate a cooperative loss such that both agents get rewarded if the language they emit allows the other agent to perform better
2.



Related Work

1. Multi-agent Communication meets Natural Language… (Angeliki)
    a. Start with a language model trained on generic (not task-specific) data and then fine-tune it with self-play in an environment
    b. They introduce a new way of combining RL updates with supervised learning by reranking language model samples
    c. Try to focus on the problem of language drift and measures that can detect language drift
    d. Main difference from this work: can we focus on two language agents that learn together?
    e. Setup: communication game between a speaker and a listener, where there is some amount of information asymmetry: one agent can see an image and has to describe a target in the image, the other agent has to take in the generation and select the target
    f.



Formulating a problem with information asymmetry & need for a cooperative loss

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1. If two agents are both language agents (inputs and outputs are over text sequences)
    a. Question answering: here one agent can learn to ask intelligent questions about a topic while the other agents learns to correctly answer the question?



This project

Confidential - DeepMind FTEs

